Indian Institute of Technology Indore

Science & Technology Council

IITISoC 2026 — Project Proposal

Selected Problem Statement (Preference 3):

Pathway-Powered Agentic Gateway

Domain: Artificial Intelligence & Machine Learning **Difficulty:** Advanced **Team Size:** 4–6

1. Team Details

1.1 Team Leader

Name	ROHITH M S
Roll Number	250005038
Discord	discord.com/users/1497592055764090964
GitHub	github.com/rohithsuresh21
LinkedIn	linkedin.com/in/rohith-m-s-964511378
Skills	PyTorch, Neural Networks, RAG, Transformers, LLMs, Feature Engineering, Time-Series Analysis

1.2 Other Team Members

#	Name	Roll Number	Key Skills
2	Santosh C	250002021	Neural Networks, GANs, LLM, Transformers
3	Adwaith P K	250003009	LLM, Transformer
4	Abhinav C	250003004	LLM, Transformer, Neural Networks
5	Sidhanth M	250005045	Web Development, JavaScript Frameworks
6	Manas Dhokar	250005025	LLM, Transformers

2. Problem Statement Preferences

Preference	Problem Statement	Domain
1	Personalized Mental Health Digital Twin using AI	AI / ML
2	Building Lightweight Multi-Task AI with MALORA	AI / ML
3	Pathway-Powered Agentic Gateway	AI / ML

3. Problem Understanding

Current LLM multi-agent frameworks frequently collapse under industrial production workloads due to high orchestration overhead, cascading errors, and catastrophic state drift under high concurrency. When parallel tool executions are triggered, managing volatile memory states synchronously leads to extreme latency spikes. Our objective for this advanced infrastructure track is to build a high-performance, provider-agnostic infrastructure proxy layer that untangles state tracking from the language model provider. By relying on a streaming architecture and independent background evaluation workers, the system guarantees low processing latency and high fault tolerance.

4. Proposed Solution

The project decomposes into three fundamental technological paradigms to isolate infrastructure lag from multi-agent tasks:

4.1 Pathway Streaming Backbone

We will build an incremental, stateful stream-processing engine using the *Pathway* framework. This serves as the decoupled, centralized source of truth for the system state. Every user prompt, internal model reasoning log, tool dispatch, and runtime environment feedback is appended as an independent event. Pathway processes these sessions concurrently, reducing synchronization bottlenecks and concurrent state inconsistencies. Additionally, Pathway's incremental computation model enables efficient state synchronization and scalable real-time processing across concurrent agent sessions.

4.2 Asynchronous Neural Watchdog (Baby Dragon Hatchling – BDH)

To maintain strict latency ceilings and trap infinite agent execution loops, a low-overhead out-of-band *Neural Watchdog* will be deployed as an asynchronous microservice. By sniffing the active Pathway event stream out-of-band, the BDH service monitors agent trajectories. If repetitive reasoning paths or redundant tool-calling patterns are isolated, it instantly emits a high-priority *Intervention Alert* to break the cycle without interrupting the primary HTTP request-response cycle. The watchdog service also improves system reliability by preventing resource exhaustion caused by recursive reasoning chains or stalled execution states.

4.3 Unified Provider-Agnostic Abstraction Layer

We will establish an Adapter interface pattern that normalises interchangeable model endpoints (e.g., local vLLM runtimes, OpenAI, Anthropic) under a single global configuration matrix. Variations in structural function-calling grammars are standardised via a unified JSON schema contract before events update the data stream. This abstraction layer also simplifies future integration of newer foundation models without requiring architectural modifications to the core orchestration pipeline.

5. Empirical Optimization Frontier

A core research component of this proposal involves constructing an automated evaluation matrix to map the speed-to-accuracy trade-offs across model parameters. We will systematically track:

- **Hyperparameter Mapping:** Evaluating changes in Temperature, Top-P, and token frequency penalties against tool-calling precision vs. execution speeds.
- **Telemetry Metrics:** Tracking Time to First Token (TTFT) alongside Cumulative End-to-End Task Execution Latency over sequential tool graphs.

6. Functional Product Proxy Surface

To rigorously stress-test our gateway framework under concurrency, we will deploy an industrial application proxy: a *High-Frequency Financial Monitor* or a *Real-Time Fleet Orchestrator*. This service will ingest thousands of high-frequency telemetry tracking streams simultaneously, pushing the Pathway data ingestion pipelines and BDH deadlock interception routines to their functional boundaries.

7. Technical Stack & Operational Frameworks

- **Data Streaming Engine:** Pathway Streaming Framework, Python Asyncio.
- **Model Execution/Serving:** Local vLLM streaming instances, Hugging Face Hub APIs, and provider-agnostic inference adapters.
- **API Framework & Interface:** FastAPI, WebSocket transport layers, Docker orchestration containers, and asynchronous request pipelines.
- **Operational Monitoring:** Runtime telemetry logging, event-driven execution tracing, latency profiling, and asynchronous task monitoring for debugging and reliability analysis.
- **Hardware Strategy:** Designed to support development and profiling on consumer-grade GPUs using local VS Code SSH nodes via Lightning AI.

8. Expected Deliverables

1. **Stateful Gateway Core Engine:** Production-grade, containerised infrastructure proxy utilising Pathway stream checkpoints.
2. **BDH Watchdog Microservice:** Functional, out-of-band loop-breaker producing operational intervention alerts.
3. **Empirical Ablation Report:** A complete research-grade interaction matrix mapping parameter adjustments directly against system P50/P90/P99 latency boundaries.
4. **Live Performance Dashboard:** Frontend application displaying real-time stream processing volumes and state-graphs.

This proposal is submitted for IITISoC 2026 under the
Science & Technology Council, IIT Indore.